Jun Ho Chai 蔡竣豪 /tsʰaɪ˥˩ tɕyn˥˩ xaʊ̯˧˥/

Email: junhoc94@gmail.com
Metrics: citations = 68, h-index = 4, i10-index = 3 (August 2026)

Education

PhD in Psychology | University of Nottingham Malaysia | 2016–2022
Thesis: *Developing very short computerised early vocabulary assessments in a multilingual setting for the Malay, Mandarin and English language* | Funded by Ministry of Science, Technology and Innovation (MOSTI)

BSc (Hons) Psychology | University of Nottingham Malaysia | 2013–2016
Final Year Project: *Role of memory constraint in emergence of language structure, a simulation study*

South Australian Certificate of Education (SACE) | Taylor's College, Malaysia | 2012 | ATAR:85.75

Teaching Experience

Lecturer | Sunway University, Malaysia | Jan–Dec 2025
- Taught and coordinated large undergraduate courses:
  Biological Psychology (N=175), Research Methods 2 (N=161);
- Designed lectures, tutorials, and assessments aligned with OBE and institutional policies;
- Developed examination papers, coursework assessments, rubrics, and marking schemes;
- Led FYP and research project supervision;
- Served on ethics review, moderation, and curriculum-related committees;
- Contributed to academic quality assurance and student-facing services such as open day

Research Assistant Professor (Part Time) | Chosun University, South Korea | Fall 2023
- Designed course structure, assessments, and marking scheme for *Affective Coding for Human Media* (College of Global Humanities);
- Delivered instruction in R for data analysis;
- Conducted weekly hands-on lab sessions;
- Conducted research workshops for students and staff

Lecturer (Part Time) | University of Nottingham Malaysia | Sep–Dec 2020
- Conducted 10 weekly lectures of Research Methods & Analysis

Research Positions

Research Assistant Professor | Chosun University, South Korea | Jan 2026–Present
- Develop AI-based methods for measuring children's language and social development from naturalistic caregivers–child audio and video recordings.
- Design and validate automated pipelines for extracting interpretable behavioural measures, including speech activity, vocalizations, gaze, responsiveness, joint engagement, and conversational structure.
- Coordinate research on automated language assessment, multimodal interaction, speech and voice activity detection, early word recognition, and computational modelling of language development.
- Prepare manuscripts, grant applications, conference presentations, and interdisciplinary research outputs in developmental psychology, psycholinguistics, and human-centred AI.

Postdoctoral Researcher (Research Assistant Professor) | Chosun University, South Korea | Jun 2022–Dec 2024
- Data analysis of naturalistic recordings and large-scale corpora (LENA, multimodal

interaction data);
- Data processing of early vocabulary and attention pattern from eye-tracker, IPLP, and parental reports;
- Supervised research assistants in data collection, preparation and coding.

Visiting Researcher (Remote) | Chosun University, South Korea | Oct 2021–May 2022
- Analysing naturalistic early language development using LENA recordings;
- Manuscripts preparation

Research Associate | University of Nottingham Malaysia | Jun–Jul 2021
- Conducted Exploratory Factor Analysis for UNM Equality, Diversity and Inclusion Survey

Postgraduate Researcher | University of Nottingham Malaysia | 2017–2020
- Developed Bayesian IRT-based computerised adaptive testing system for brief tablet-based language assessments in children under 3;
- Analyzed vocabulary and response data using generalized linear mixed-effects models

Publications

Peer-Reviewed Journal Articles

Chai, J. H., Zapiór, B., & Ko, E. S. (2026). Developmental Shift in Multimodal Cue Usage in Early Mother‐Child Conversational Initiation. *Infancy*, *31*(3), e70096. https://doi.org/10.1111/infa.70096

Chai, J. H., Mun, S., & Ko, E. S. (2026). The Alignment Between Language Properties and Computational Algorithms Enhances Statistical Word Segmentation: Evidence from Korean Child-Directed Speech. *Journal of Child Language*. https://doi.org/10.1017/S0305000926100646

Chai, J. H., & Ko, E. S. (2025). Input Properties Shape Word Segmentation Performance Across Child Development: A Computational Modeling Study. *Infancy*, *30*(5), e70046. https://doi.org/10.1111/infa.70046

Chai, J. H., Jung, J., & Ko, E. S. (2025). The dynamics of initiation in caregiver–child conversational interactions. *Journal of Child Language*, *52*(4), 839–860. https://doi.org/10.1017/S030500092400062X

Chai, J. H., Low, H. M., Wong, T. P., Onnis, L., & Mayor, J. (2021). Extra-linguistic modulation of the English noun-bias: Evidence from Malaysian bilingual infants and toddlers. *Journal of Cultural Cognitive Science*, *5*, 49–64. https://link.springer.com/article/10.1007/s41809-021-00078-5

Lo, C. H., Rosslund, A., Chai, J. H., Mayor, J., & Kartushina, N. (2021). Tablet assessment of word comprehension reveals coarse word representations in 18-20-month-old toddlers. *Infancy*. https://onlinelibrary.wiley.com/doi/full/10.1111/infa.12401

Chai, J. H., Lo, C. H., & Mayor, J. (2020). A Bayesian-Inspired Item Response Theory–Based framework to produce very short versions of Macarthur–Bates Communicative Development Inventories. *Journal of Speech, Language, and Hearing Research*, *63*(10), 3488–3500. http://dx.doi.org/10.1044/2020_JSLHR-20-00361

Peer-Reviewed Proceedings

Chai, J. H., McDonald, M., & Ko, E. S. (2025). Looking beyond parental reports: systematic biases in early word recognition assessment. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47). https://escholarship.org/uc/item/7511s972

Kim, M., Shim, M., Chai, J. H., Ko, E. S., & Lee, Y. (20225). Lookee: Gaze Tracking-based Infant Vocabulary Comprehension Assessment and Analysis. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (pp. 1–15). https://doi.org/10.1145/3706598.3713386

PhD Thesis

Chai, J. H. (2022). *Developing very short computerised early vocabulary assessments in a multilingual setting* (Doctoral dissertation, University of Nottingham). https://eprints.nottingham.ac.uk/67007/

Manuscripts

Jung, S., Chai, J. H.*, Buhnila, I., & Ko, E.-S. Classifying child-directed speech and shared book reading from LENA: Language-specific modeling and temporal resolution effects. Manuscript under review. *Co-first author (equal contribution)

Chai, J. H., Low, H. M., Wong, T. P., & Mayor, J. The interplay of socio-economic status and language exposure in shaping bilingual vocabulary development in Malaysian children. Manuscript in preparation.

Chai, J. H., Zapiór, B., & Ko, E.-S. Developmental shift in multimodal cue usage in early mother-child conversational initiation. Manuscript under review.

Grants

ManyBabies 5 Grant (Co-PI, with Woo Pei Jun @ Sunway University, Malaysia) – USD 2,500

- Co-leads the local implementation of an international infant looking-time study.
- Secures ethics approval, supervises recruitment, and trains research assistants.
- Oversees data collection and quality control in line with consortium protocols.
- Leads statistical analysis for local data and contributes to cross-site models.
- Co-authors consortium manuscripts and coordinates reporting to the funder.

HK 3.0: Language, Cognition, and Society from the National Research Foundation of Korea (Collaborator, with PI: Professor Eon-Suk Ko @ Chosun University, South Korea) – USD 3,500,000

- Lead projects on early language development and elderly speech with an ecological valid data-driven approach.
- Develop annotation and analysis pipelines for caregiver–child interaction data and elderly speech.
- Mentor HK 3.0 researchers and contribute to publications and symposia.

Academic Service

Founding member and steering board member, DARCLE-Asia

- A regional network advancing long-format recordings and naturalistic developmental data across Asia and Oceania.

Board member of LangView

- International consortium on language variation in children's input environments and developmental outcomes.

Peer Review

- *Current Psychology*; *Journal of Child Language*; *Infant and Child Development*

Invited Talks and Workshops

8 May 2026 | Who Initiates and How It Happens Matters: Conversational Dynamics and Early

Language Development (online). Psycholinguistics Lab Group, University of Birmingham, UK

15 April 2026 | Who Initiates and How It Happens Matters: Conversational Dynamics and Early Language Development (online). DARCLE: Daylong Audio Recordings of Children's Linguistic Environments.

22–23 Jan 2026 | Optimising Gaze Feature Selection for Evidence of Infant Word Recognition Using Random Forests (online). 3rd POSTECH MINDS-Chosun CDS Joint Workshop: Human Language Acquisition through Machine Learning and AI, Jeju Island, South Korea.

13 Sep 2025 | Data analysis with LENA-generated files. HK 3.0 Workshop: Language, Cognition, and Society: A Data-Driven Lifespan Perspective. Center for Data Science in Humanities, Chosun University, South Korea.

11 Sep 2025 | Convergence Between Eye-Tracking-Based Word Recognition and Parental Report in Assessing Language Development in 14-Month-Old Korean Infants. HK 3.0 Workshop: Language, Cognition, and Society: A Data-Driven Lifespan Perspective. Center for Data Science in Humanities, Chosun University, South Korea.

25 Jun 2024 | Investigating the Convergence of Child Language Assessment Measures with 14-month-old Korean Infants. Chosun Center for Data Science in Humanities Annual Workshop, Chosun University, South Korea.

25 Jun 2024 | Accessing and analyzing Peekbank's eye-tracking datasets. Chosun Center for Data Science in Humanities Annual Workshop, Chosun University, South Korea.

24 Jun 2024 | R workshop based on behavioural data. Chosun Center for Data Science in Humanities Annual Workshop, Chosun University, South Korea.

30 Jan 2024 | Refining Infant Word Recognition Analysis through Optimised Gaze Feature Selection based on Random Forest. Chosun University, South Korea.

25 Jan 2024 | Going beyond reporting of main effects. 2023 Experimental Phonetics Research Association Winter Workshop, Hankuk University of Foreign Studies, South Korea.

27–29 Dec 2022 | Basic and intermediate statistics using R. Chosun University, South Korea.

7 & 29 Apr 2020 | Introduction to doing statistics in R. University of Nottingham Malaysia.

5 Dec 2019 | Research Workshop - Analysing Quantitative Surveys. University of Nottingham Malaysia.

Conference Presentations

Chai, J. H., Wong, T. P., Low, H. M., Mayor, J., (2026). The Interplay of Socioeconomic Status and Diverse Language Environments in Early Bilingual Vocabulary Development in Malaysia. In *Many Paths to Language Conference 2026*, Nijmegen, Netherlands.

Chai, J. H., Jung, S., Buhnila, I., & Ko, E. S. (2026, July). Automated Classification of Child-Directed Speech and Shared Book Reading from LENA Recordings in Korean. In *The International Congress of Infant Studies (ICIS)*.

Chai, J. H., & Ko, E-S. (2026). Human-speech detection in naturalistic child-centered audio: An interpretable open-source ensemble voice activity detection approach. *SICOL 2026*, Seoul, South Korea.

Chai, J. H., Jung, J., & Ko, E.-S. (2025). *A LENA-based study of gendered input in Korean*

parent–child interactions. Poster presented at the 50th Annual Boston University Conference on Language Development (BUCLD 50), Boston University, Boston, MA, USA.

Ko, E.-S., Chai, J. H., McDonald, M., & Jo, J. (2025). *Age-related changes in sound symbolism and word learning in Korean infants*. Poster presented at the 50th Annual Boston University Conference on Language Development (BUCLD 50), Boston University, Boston, MA, USA.

Chai, J. H., Mun, S., & Ko, E. S. (2025). Dissecting the interplay between corpus properties, algorithm, and word segmentation performance. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).

Chai, J. H., McDonald, M., & Ko, E. S. (2025). Looking beyond parental reports: systematic biases in early word recognition assessment. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).

Kim, M., Shim, M., Chai, J. H., Ko, E. S., & Lee, Y. (2025, April). Lookee: Gaze Tracking-based Infant Vocabulary Comprehension Assessment and Analysis. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (pp. 1–15).

Chai, J. H., Kim, M., Shim, M., Lee Y., & Ko, E-K. (2024). Refining Infant Word Recognition Analysis through Optimised Gaze Feature Selection based on Random Forest. *ICIS 2024 Congress*, Glasgow, UK.

Chai, J. H., Zapiór, B., & Ko, E-K. (2024). How do mothers and children initiate conversational exchanges?: The dynamics of multimodal cue usage in beginning vocal exchanges across child development. ICIS 2024 Congress, Glasgow, UK.

Chai, J. H., McDonald, M., & Ko, E-K. (2024). Investigating the Convergence of Child Language Assessment Measures in 14-month-old Korean Infants.

Chai, J. H., Mun, S., & Ko, E. S. (2023). How Child-Directed Speech Enhances Word Segmentation: Insights from a Computational Modeling Study of Korean. In *The 48th Boston University Conference on Language Development*, USA.

Jung, J., Chai, J. H., & Ko, E. S. (2023). Socioeconomic Status, Parental Play and Book-reading, Maternal Work Status and Vocabulary Development in Young Korean Children. In *The 48th Boston University Conference on Language Development*, USA.

Chai, J. H., Mun, S., & Ko, E. S. (2023). How caregivers' speech patterns enhance word segmentation: Insights from a computational modelling study on a Korean corpus. In *Many Paths to Language (MPaL) 2023*, Netherlands.

Jung, J., Chai, J. H., & Ko, E. S. (2023). Socioeconomic status, parental play and book-reading, maternal work status and vocabulary development in young Korean children. In *Many Paths to Language (MPaL) 2023*, Netherlands.

Chai, J. H., Jung, J., Kwon, T., & Ko, E. S. (2022). Mother's initiative role in conversational exchanges and its effect on children's language outcome. In *XXIII International Congress of Infant Studies 2022 Developmental Cascades*, Canada.

Jung, J., Chai, J. H., & Ko, E. S. (2022). The effect of socioeconomic status on early vocabulary size in South Korean children. In *XXIII International Congress of Infant Studies 2022 Developmental Cascades*, Canada.

Mayor, J., Chai, J. H., Lo, C. H., & Mani, N. (2020). Babylex-IRT: Estimating early vocabulary sizes using a Bayesian-inspired item-response theory approach. In *The International Congress of Infant Studies 2020*, Canada.

Awards

2022 | The LEGO Foundation Scholar Travel Awards for ICIS 2022 (unclaimed due to COVID-19 VISA delays)

2018 | Postgraduate Cross Cultural Travel Award, School of Psychology, University of Nottingham Malaysia. Travelled to University of Oslo for research purposes.

Skills

Data Analytics: R, Python, UNIX, SPSS, Matlab, Praat, CLAN, ELAN
Experimental Design: PsychoPy
AI Generative Tools: Perplexity, Kiro, ChatGPT (Codex)
AI Annotation Tools: WebRTC VAD, Silero, RNNnoise, Kaldi, Pyannote, Whisper, DeepLabCut, Mediapipe, OpenFace, OpenPose
Languages: Mandarin (Native), Cantonese (Native), English (Bilingual), Malay (Professional), Korean (Limited Working)

References

Prof. Dr. Eon-Suk Ko
Professor
Department of English Language and Literature
Director of Center for Data Science in Chosun University
Chosun University
Republic of Korea
+82-62-230-6551
eonsukko@chosun.ac.kr

Prof. Dr. Julien Mayor
Professor
Department of Psychology
University of Oslo
Norway
+47 22845149
julien.mayor@psykologi.uio.no

Dr. Tze Peng Wong
Associate Professor,
Director of Teaching & Learning, Faculty of Social Sciences,
Faculty of Social Sciences
University of Nottingham Malaysia
Malaysia
+6 (03) 8725 3590
TzePeng.Wong@nottingham.edu.my